Before leaving from the lab, make sure batteries and controller have charge.

Before you fly check if there are any restrictions to fly in the area you are flying. You can check it in air control app. Add your license number in the app.

Making drone ready to fly

1. **Set up ground control points for better GPS:**

- Connect wires between ipad and the stand.
- With a controller, turn on the hotspot and connect WIFI (drone serial no.) to GCP's.
- This helps to set up ground map as well (later on).

2. **Setting up Done:**

- Drone box has 6 latches. To open them, press and lift
- Take drone out of the box
- **Flip** the drone upside down. Pull and slide the **bigger** knobs.
- Pull the landing gear same way
- Extend blades, make the gps antenna upright by pulling up knobs on both sides together
- Fit the camera by sliding it on the side where arrow points. **Turn the red circular knob** towards right,
- Water color box has a camera **cable**, ssd card holder and other precious items. One green camera cable is stored in that box.
- Take out green cable and Plug 2 cables in camera by pressing them down (you will hear a clicking sound while connecting them). One end of green cable has spot to plug towards down front of the drone.

- Check if SD card is all the way inside in the camera.
- Take out the **battery** with green rubber from the box. Slide it. You can see a pad if you are confused which end to slide.
- Plug in the battery cable. Make sure its all the way plugged in. Put straps over the battery
- Drone is turned on now. Wait until it beeps.
- Open **wifi** in your phone and connect rededge wifi (drone wifi). Open 192.168.10.254 path in safari/google. In the home section, if battery or sd card are not available, it will show red warning. If that's good, Go to **settings sign,** basic configuration, auto capture mode to **Overlap** (80%), and make sure your flight height is same there. Click to camera icon and click streaming. There are also buttons in front of the camera and front of drone to click pictures if you can't connect to wifi. (PS: if there is no storage in the sd card, from setting, storage section, reformat it only if you know the material in the sd card is already backed up in the lab).
  Set up: usually overlap % as 80 %, altitude as 25-30 meters, that will automatically adjust speed of drone in controller part (not shown here)
- Take out **calibration** panel. Put it in sun near field with grey side clearly visible to sun (barcode side is not imp.). Hold the drone facing opposite to the sun and take pictures of calibration panel in the path above. DO NOT block panel and make sure there is no shadow.

- Generally, one person holds the drone and another person take pics when full calibration panel is seen properly (if you are using phone to click pictures).
- Check

Controller part

- Straighten the antennas
- Open the flight deck applicqtion
  Make flight map
- Click on sign A~B (symbol for mapping)
- Click pattern, then survey and move four points at corner of the field. You can also walk to the corners of the field to mark exact corners. Deck sign shows where you/controller is
- After you click survey, you find option to put altitude (generally 25 m), angle (90º /0º) on the right side of the screen. We want drone to fly on the longer side of the field. Put 0º degree if the drone and flight path are facing in the same direction and vice-versa
  Set up: usually overlap % as 80 %, altitude as 25-30 meters, that will automatically adjust speed of drone in controller part.
- Make sure you click optimize speed
- Then save file with unique name
- If you make any changes after saving, save again

➢ **After you make flight map/if flight map is already made**, click on sign A~B, open your flight map, then **upload** mission.

➢ Click on deck sign, you will see option to slide to continue mission. **Do not slide** it now.

Lets' fly now

- Move the left thumb stick **down and right** to arm the drone. There is sticker on the controller which says which way is arm and which way is disarm. Arm means making the drone ready and disarm means stopping the drone.
- Once it says armed, shift the left thumb stick up/front until the drone reach certain height.
- Move the **landing** gear up. You can do it by shifting the gear sign up in left side of controller.
- Slide '**Continue** the mission' option in controller.

- Then monitor for a while if it is actually following the right track and you haven't uploaded wrong map by mistake. In such case, single click to stop and double to make it to manual and then bring it back, land , and then start again with correct flight map. (DON'T PANIC, usually single click will stop it so it won't run away on its own..).

Land the drone

- After the drone completes mission, it says mission completed, then we need to change it to manual mode to land it back
- Press the **manual button in right edge of the controller twice**. When you press once, it breaks flight mode. Pressing second time changes the mode to manual.
- Use right thumb stick to move drone in a safe place (usually outside of the main plots/in grass)
- Right thumb stick helps you to move back and forth, left and right. Pressing it towards front moves the drone front. See which side is the drone facing to determine its front and back side.
- Touch the **gear** symbol in deck app. Landing gear comes down.
- There is also **RTL button** which you can click/touch in controller and swipe to continue it (similar way when you continue mission). This will bring back the drone to original place where we launched the flight. This option can bring the drone back to the ground where we started. But this might be risky as the land surface may not be even and drone may try to land at a faster speed and break its landing gears. When you use this option, break the RTL mode when it **comes above you** (like 10-12 feet from the ground). Break it by pressing the same manual button (mentioned above). Press twice to bring it to manual mode.
- Touch the gear symbol in deck app. Landing gear comes down.
- Then your drone is ready to land. Shift left thumb stick down (not all the way) till the drone touches the ground. Do not release the thumb stick. Continue pressing it down. Make sure you press the stick slightly for a safe and slow landing.
- When drone is in the ground, move the left thumb stick all the way down and then left to disarm it.
- The drone is disarmed
- Take pictures of calibration panel

Putting drone back to box

- Follow the steps for "Making drone ready" in opposite direction
- Plug out the **battery** cables. Take off the battery straps. Put the battery back in the box with red rubber band
- Press down the **camera cable** clips to release them. Put the green cable back in plastic box.

- Take out the **camera**. To take it out, turn the red circular **knob** towards left. Then hold the blue/red surface (which is holding the camera) with one hand and slide the camera in opposite direction of the arrow
- Put the camera with its cover in the drone box
- Shift the gear **knobs** & blade knobs by pulling out and sliding.
- Slide down the antenna by pressing the knobs from both sides
- Put the drone back in the box.
- Check if all the equipments are in the box (calibration panel, camera, battery, drone)
- Put the latches of drone box

Take it back to storage location and keep battery, controller, ipad and any electronic components on charging station.